

“Where AI can make an impact, where it can bring equity, where it can bring change in the lives of people is more important to me as a policymaker. This is what we will stress on”

“The Tatas and Telangana government have put in money together to modernise our industrial training institutes and put robotics as a core subject”

D. SRIDHAR BABU
Minister for Information Technology, Electronics & Communications, Telangana

Did you know that up to 113 unique components go into making a bionic hand that can replicate the functions of a human hand? Or that robotic systems in Indian factories can make up to eight cars on a single assembly line? Or that 250 robotic systems across Indian hospitals are currently assisting in surgeries? The INDIA TODAY Robotics and AI Conclave in Hyderabad on November 6 covered such fascinating developments in the tech space, and united policymakers, innovators and academics who shared their insights on the sweeping changes this is set to bring in the health, manufacturing and agriculture sectors.

“We are entering a world of hyper specialisation,” Rajiv Gupta, managing director and senior partner at Boston Consulting Group, said. “You are almost certain that 1.5 million-odd jobs out of [India’s 8 million-strong tech workforce] will get displaced,” he said. “But there are new jobs getting created already... in tech companies, banks, consumer good companies, manufacturing,” he added, referring to roles that involve prompting and operating AI programs. Jaspreet Bindra, managing director and founder, Tech Whisperer, added, “All of us don’t need to be AI experts, but we all need to be literate in it, the same way you become literate in English to become a successful professional.” Dr John Bruce, director of placements & corporate affairs at the Sathyabama Institute of Science & Technology, said a collaborative workflow between humans and machines will soon emerge.

One Indian state leading the charge is Telangana. According to D. Sridhar Babu, the state’s minister for IT and Industries, 65 industrial training institutes in Telangana are being upgraded with a focus on robotics. Two arenas in which it is already proving useful are health and manufacturing. According to Dr Sanjay Addla, senior consultant in uro-oncology and robotic surgery at Apollo Hospitals, while robot-assisted surgeries have been around for decades, innovations could deliver medical services to under-served regions. Pointing to dark factories—manufacturing units run entirely with robots—Gopalakrishnan C.S., chief manufacturing officer of Hyundai Motor India, remarked on the incoming paradigm shift. Two startups hailing it are Makers Hive, with a cost-effective prosthetic arm, and Harvested Robotics, with its tractor-mounted weed-zapping laser.

Autonomous systems are already in use in defence. “Tomorrow, the most difficult job of soldiers will be taken away,” said Dr Narayan Panigrahi, Scientist ‘H’ and group head, GIS, of the Centre for Artificial Intelligence & Robotics. Transparency will be important as more sophisticated machines emerge, according to Dr L. Lakshmanan, professor at Sathyabama. Policy, incentives and greater R&D must align, as Kisshhan P.S.V., chairman of the All India Robotics Association, pointed out. The future’s bright, it needs to be safer too. ■